

VORTEX-GEN — FULL PROJECT ARCHITECTURE
Generated: 5/3/2026

backend

```
|-- .env.example
|-- Dockerfile
|-- package-lock.json
|-- package.json
|-- requirements.txt
|-- run_nf.py
+-- server.js
```

frontend

```
|-- public
|   |-- banner.png
|   |-- favicon.svg
|   |-- icons.svg
|   +-- vortex-gen-logo.png
+-- src
    |-- assets
    |   |-- images
    |   |   |-- loading
    |   |   |   +-- loading-base.png
    |   |   |-- aircraft-topview.png
    |   |   |-- hero.png
    |   |   +-- logo.png
    |   |-- aircraft-topview.png
    |   |-- hero.png
    |   |-- logo.png
    |   |-- react.svg
    |   |-- vite.svg
    |   |-- vortex_gen_loading_screen_base.png
    |   +-- vortex_gen_loading_screen_start.png
    |-- components
    |   |-- layout
    |   |   |-- LoadingScreen.css
    |   |   +-- LoadingScreen.tsx
    |   |-- ui
    |   |   |-- Button.tsx
    |   |   |-- ControlSlider.tsx
    |   |   |-- Modal.tsx
    |   |   |-- Slider.tsx
    |   |   +-- Spinner.tsx
    |   |-- AeroFactsPanel.jsx
    |   |-- AuthGuard.jsx
    |   |-- ControlSlider.jsx
    |   |-- DataChart.jsx
    |   |-- Export3DModal.jsx
    |   |-- LoadingScreen.css
    |   |-- LoadingScreen.jsx
    |   |-- PdfReportTemplate.jsx
    |   |-- PolarChart.jsx
    |   |-- ShapeCard.jsx
    |   +-- SimulationView.jsx
    |-- config
    |   +-- constants.ts
    |-- context
    |   +-- AppContext.jsx
    |-- features
```

```
| -- academy
| | -- components
| | | +-- AeroFactsPanel.tsx
| | -- layout
| | | +-- AcademyLayout.tsx
| | -- pages
| | | -- AirfoilPage.tsx
| | | -- ExplorerPage.tsx
| | | -- FuselagePage.tsx
| | | +-- WingsPage.tsx
| | +-- index.ts
| -- auth
| | -- components
| | | +-- AuthGuard.tsx
| | -- pages
| | | -- LoginPage.tsx
| | | +-- SignupPage.tsx
| | +-- index.ts
| -- lab
| | -- components
| | | -- charts
| | | | -- DataChart.tsx
| | | | +-- PolarChart.tsx
| | | -- export
| | | | -- Export3DModal.tsx
| | | | +-- PdfReportTemplate.tsx
| | | +-- simulation
| | | | -- ShapeCard.tsx
| | | | +-- SimulationView.tsx
| | -- data
| | | -- envPresets.ts
| | | +-- shapes.ts
| | -- hooks
| | | -- useAerodynamics.ts
| | | +-- useSimulation.ts
| | -- layout
| | | +-- LabLayout.tsx
| | -- pages
| | | +-- DashboardPage.tsx
| | +-- index.ts
| -- landing
| | -- components
| | +-- pages
| | | +-- LandingPage.tsx
| -- pricing
| | +-- pages
| | | +-- PricingPage.tsx
| -- profile
| | +-- pages
| | | +-- ProfilePage.tsx
+-- settings
| | -- components
| | +-- pages
| | | +-- SettingsPage.tsx
-- hooks
| | -- useAppInit.ts
| | -- useBackendStatus.ts
+-- useTheme.ts
```

```

|-- i18n
|   |-- locales
|   |   |-- ar
|   |   |   |-- academy.json
|   |   |   |-- common.json
|   |   |   +-- lab.json
|   |   +-- en
|   |       |-- academy.json
|   |       |-- common.json
|   |       +-- lab.json
|   +-- index.ts
|-- layouts
|   |-- DashboardLayout.jsx
|   +-- ExplorerLayout.jsx
|-- lib
|   |-- apiClient.ts
|   |-- supabaseClient.ts
|   +-- utils.ts
|-- pages
|   |-- explore
|   |   |-- AirfoilSection.jsx
|   |   |-- FuselageSection.jsx
|   |   |-- TailSection.jsx
|   |   +-- WingsSection.jsx
|   |-- lab
|   |   |-- FuselageLab.jsx
|   |   |-- LabHub.jsx
|   |   |-- TailLab.jsx
|   |   +-- WingsLab.jsx
|   |-- Explorer.jsx
|   |-- Home.jsx
|   |-- LandingPage.jsx
|   |-- Login.jsx
|   |-- Pricing.jsx
|   |-- Profile.jsx
|   |-- Settings.jsx
|   +-- Signup.jsx
|-- services
|   |-- apiService.js
|   +-- supabaseClient.js
|-- store
|   |-- appContextShim.ts
|   |-- authStore.ts
|   |-- index.ts
|   |-- settingsStore.ts
|   +-- simulationStore.ts
|-- styles
|   |-- themes
|   |   |-- dark.css
|   |   +-- light.css
|   |-- animations.css
|   +-- index.css
|-- App.css
|-- App.jsx
|-- App.tsx
|-- index.css
|-- main.jsx
+-- main.tsx

```

.env.example
.gitignore
MIGRATION_COMPLETE.md
README.md
eslint.config.js
index.html
package-lock.json
package.json
vite.config.js